

PROFESSIONAL INSTALLATION MANUAL

Ubuntu Server Installation and Configuration

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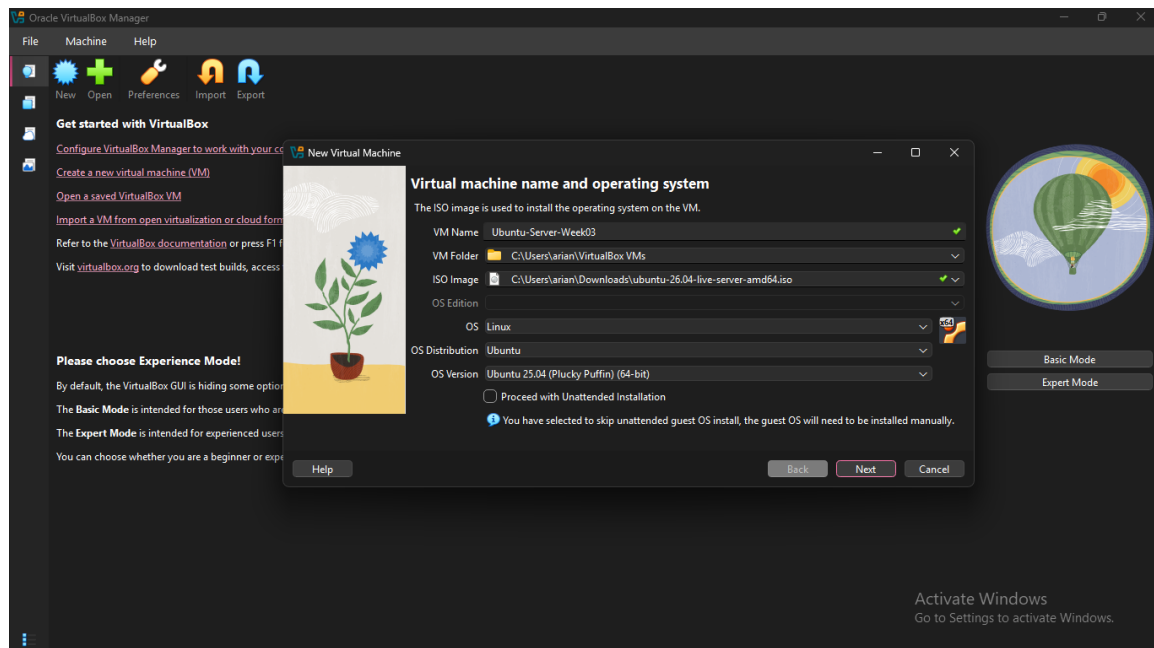
1. Introduction

This manual provides step-by-step instructions for installing, configuring, and verifying Ubuntu Server in a virtual machine. It is designed so that another system administrator can follow the procedure without needing additional instructions. The manual covers installation, basic server configuration, verification, troubleshooting, and recommended best practices.

2. Ubuntu Server Installation Guide

2.1 Create the Virtual Machine

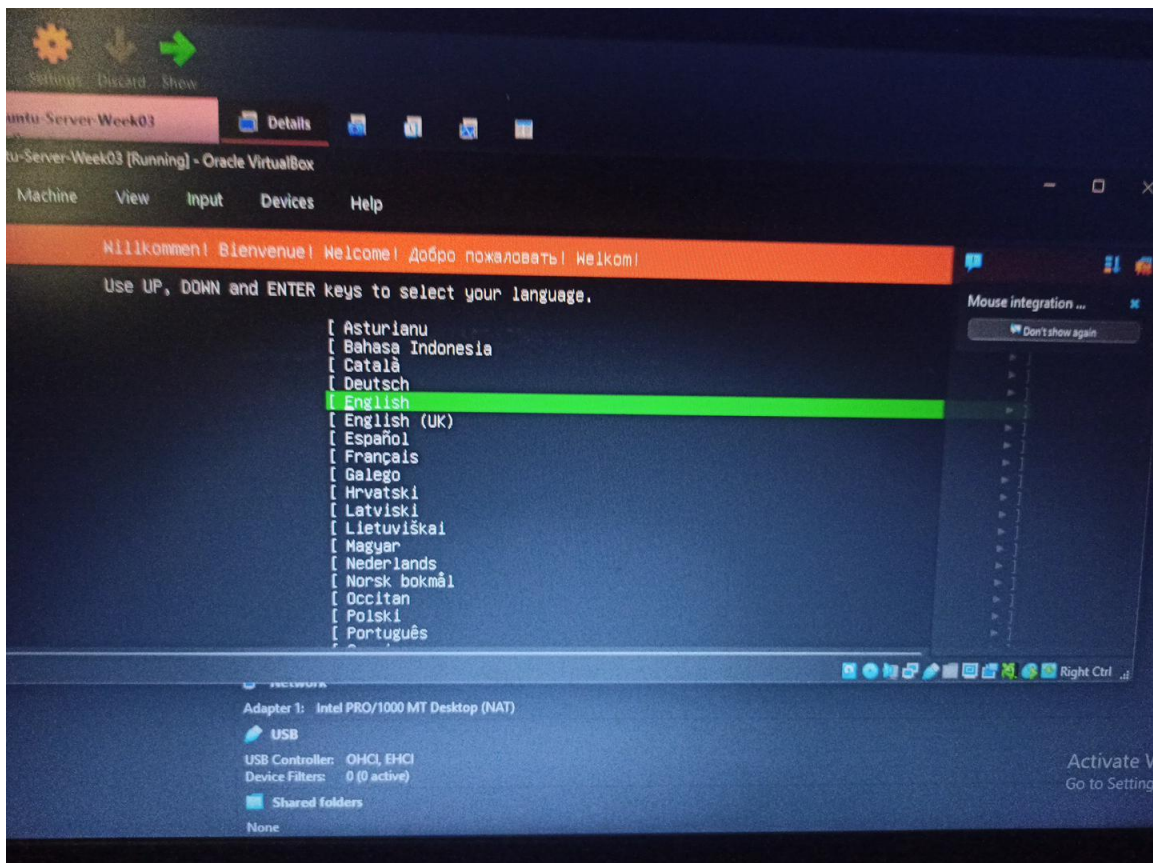
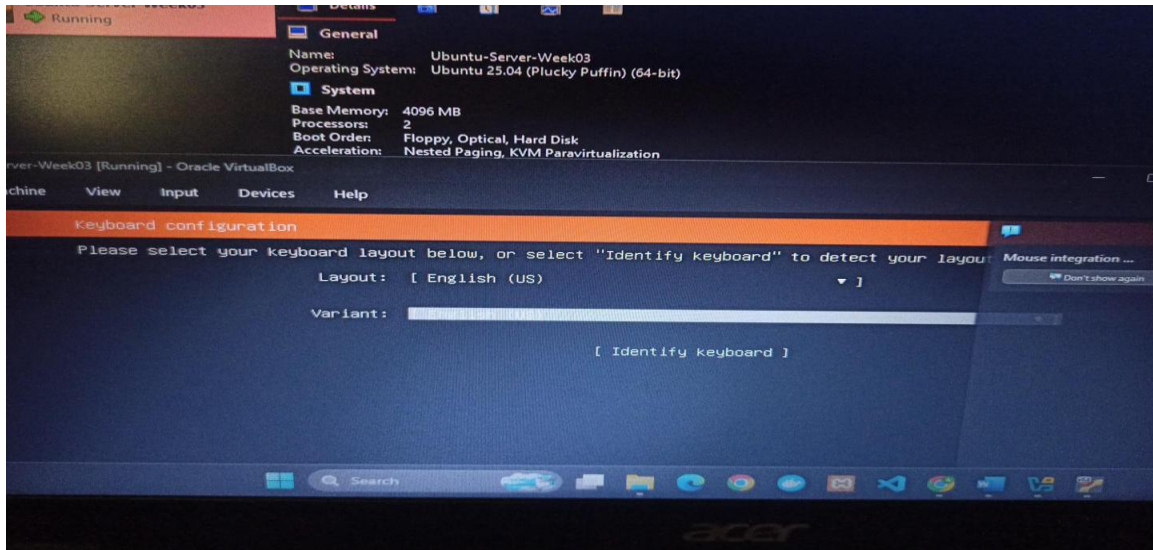
1. Open Oracle VM VirtualBox.
2. Click New.
3. Enter the name: Ubuntu Server.
4. Select the Ubuntu Server ISO.
5. Set the required RAM.
6. Set the required CPU.
7. Create a virtual hard disk.
8. Set the network adapter to NAT.
9. Click Finish.

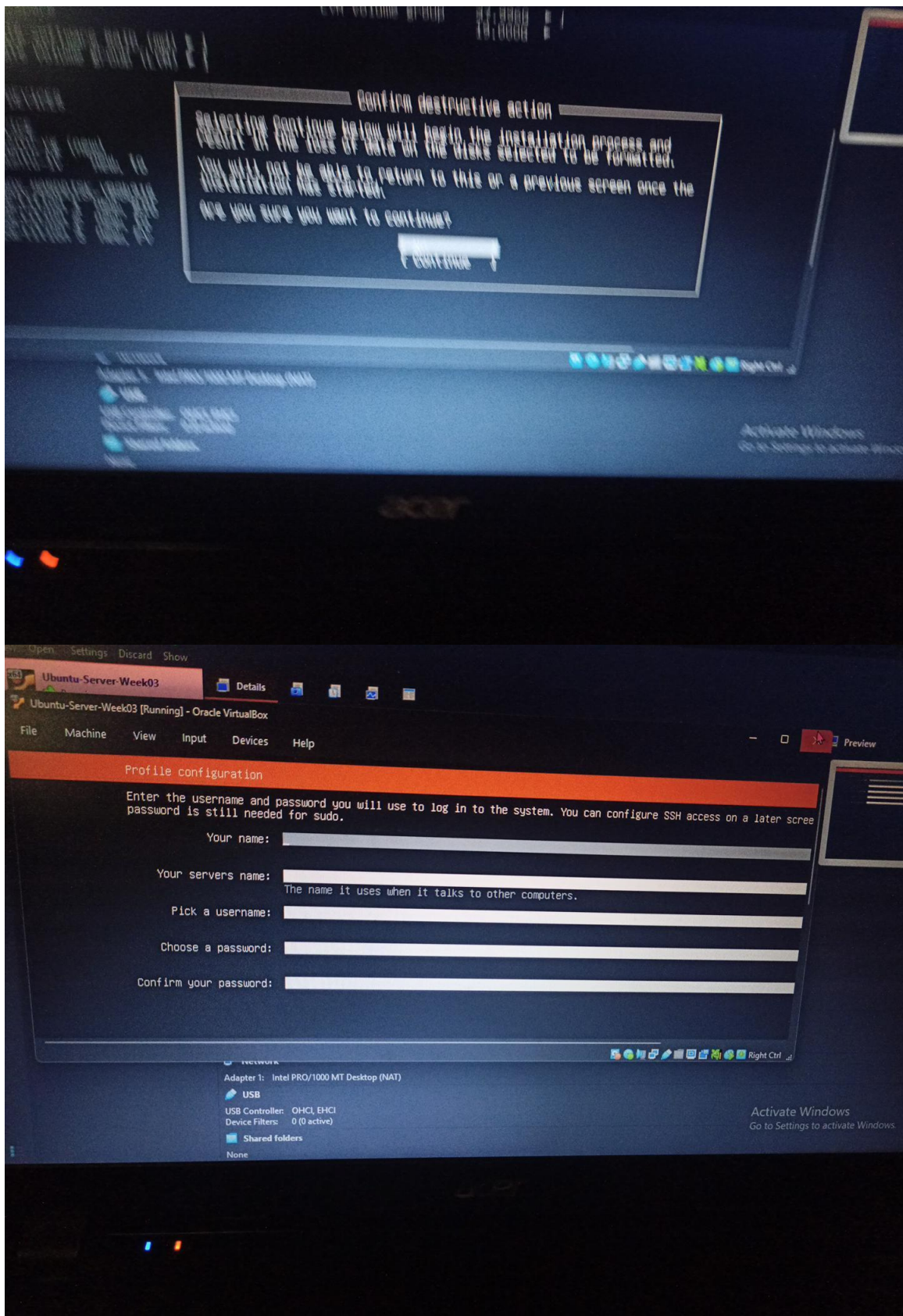


2.2 Start the Installation

1. Select the Ubuntu Server virtual machine.
2. Click Start.
3. Wait for the Ubuntu installation screen.
4. Select English.

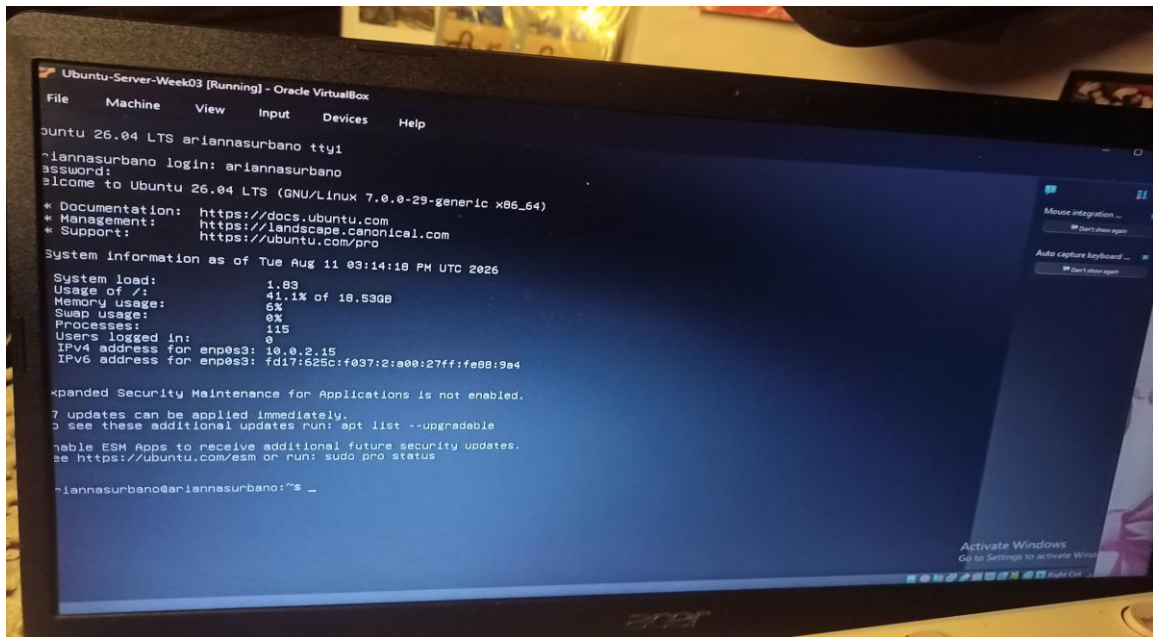
5. Continue through the installation options.
6. Create the server user account.
7. Create a password.
8. Complete the installation.
9. Restart the virtual machine when requested.





2.3 Login

After installation, enter the username and password created during setup, then press Enter. The Ubuntu Server command-line interface should appear.



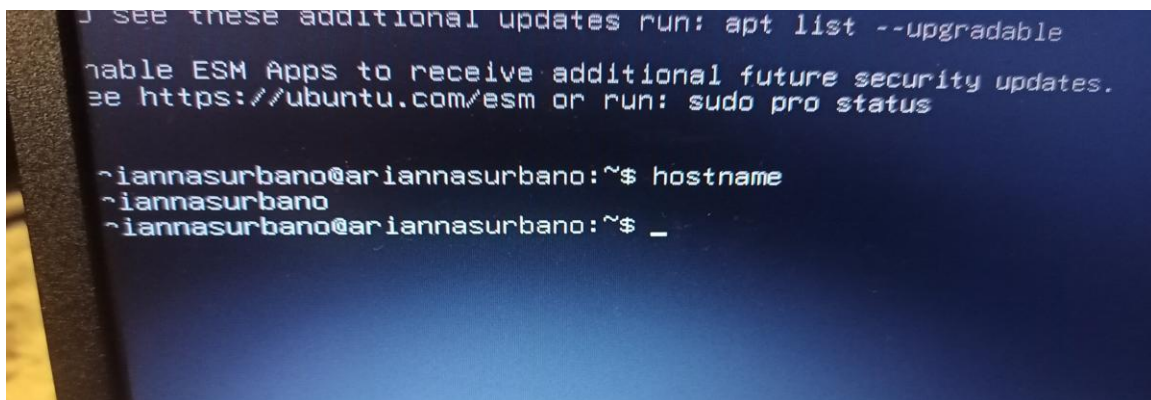
3. Initial Server Configuration

3.1 Verify Hostname

Run:

hostname

Expected result: ariannasurbano

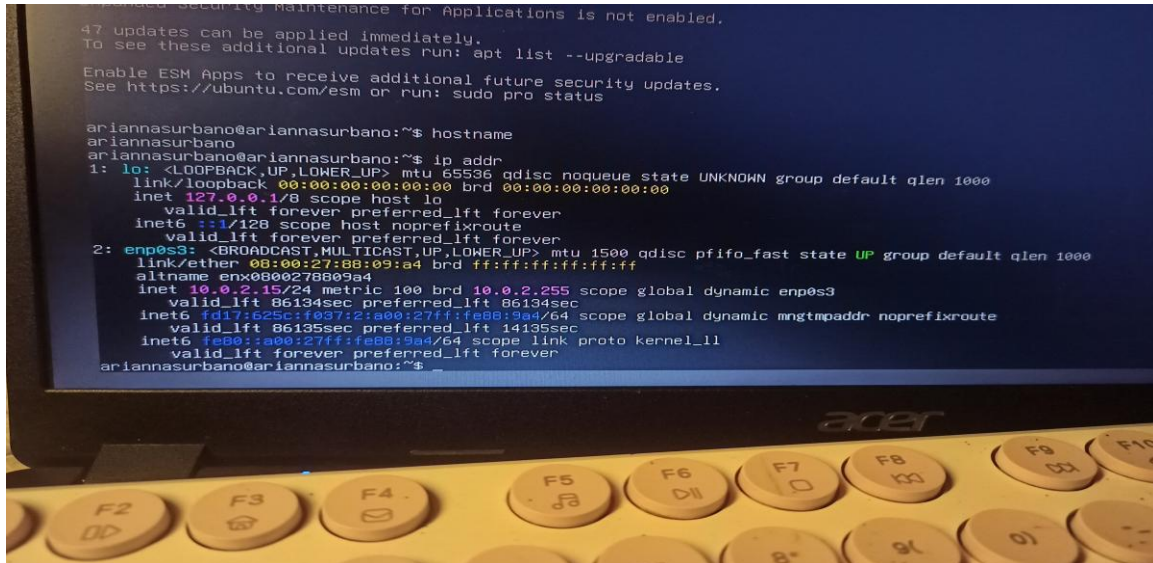


3.2 Check IP Address

Run:

ip addr

The IP address used in the activity was 10.0.2.15.



A photograph of a laptop screen displaying a terminal window. The terminal shows the output of the 'ip addr' command. At the top, there are system update notifications. Below, the terminal shows the user's prompt and the command 'ip addr'. The output lists network interfaces: 'lo' (loopback) with IP 127.0.0.1 and 'enp0s3' (ethernet) with IP 10.0.2.15. The IP address 10.0.2.15 is highlighted in green. The terminal is running on a system named 'ariannasurbano'.

```
47 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

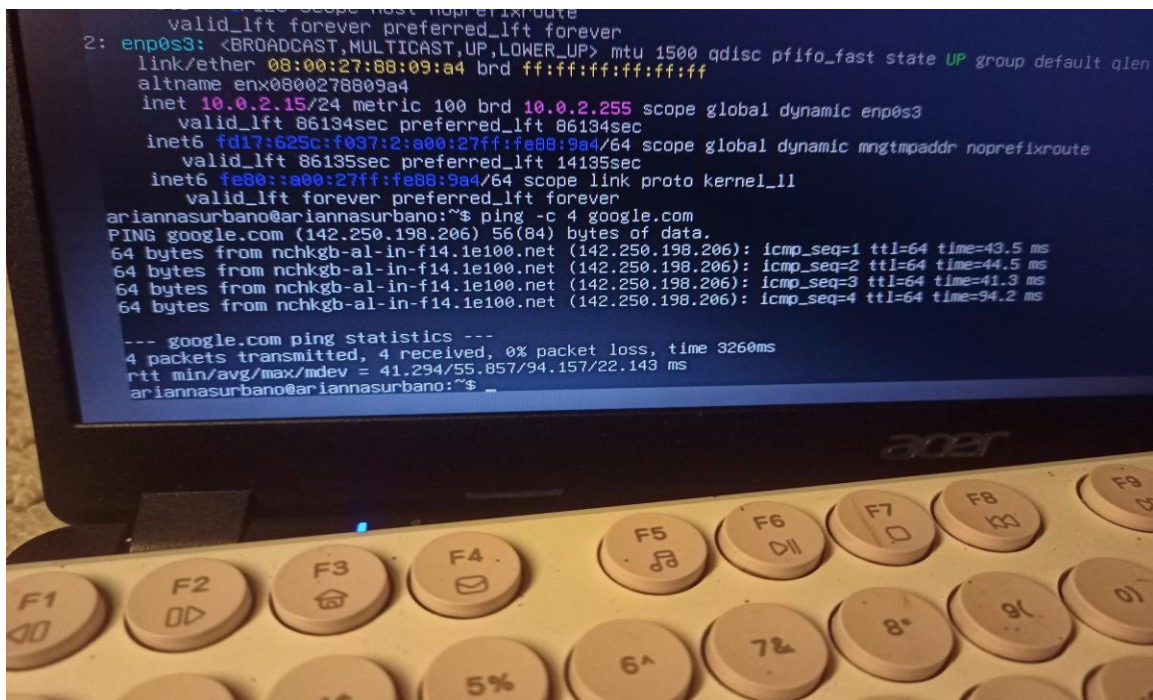
ariannasurbano@ariannasurbano:~$ hostname
ariannasurbano
ariannasurbano@ariannasurbano:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:88:09:a4 brd ff:ff:ff:ff:ff:ff
    altname enx0800278809a4
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86134sec preferred_lft 86134sec
    inet6 fd17:625c:f037:2:a00:27ff:fe88:9a4/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86135sec preferred_lft 14135sec
    inet6 fe80::a00:27ff:fe88:9a4/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
ariannasurbano@ariannasurbano:~$
```

3.3 Test Internet Connection

Run:

ping -c 4 google.com

A successful result should show replies from Google with no packet loss.



A photograph of a laptop screen displaying a terminal window. The terminal shows the output of the 'ping -c 4 google.com' command. It displays four successful ping replies from Google, each with a time response. Below the replies, it shows 'ping statistics' indicating 4 packets transmitted, 4 received, 0% packet loss, and a total time of 326ms. The terminal is running on a system named 'ariannasurbano'.

```
ariannasurbano@ariannasurbano:~$ ping -c 4 google.com
PING google.com (142.250.198.206) 56(84) bytes of data.
64 bytes from nchkgb-al-in-f14.1e100.net (142.250.198.206): icmp_seq=1 ttl=64 time=43.5 ms
64 bytes from nchkgb-al-in-f14.1e100.net (142.250.198.206): icmp_seq=2 ttl=64 time=44.5 ms
64 bytes from nchkgb-al-in-f14.1e100.net (142.250.198.206): icmp_seq=3 ttl=64 time=41.3 ms
64 bytes from nchkgb-al-in-f14.1e100.net (142.250.198.206): icmp_seq=4 ttl=64 time=34.2 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 326ms
rtt min/avg/max/mdev = 41.294/55.857/94.157/22.143 ms
ariannasurbano@ariannasurbano:~$
```

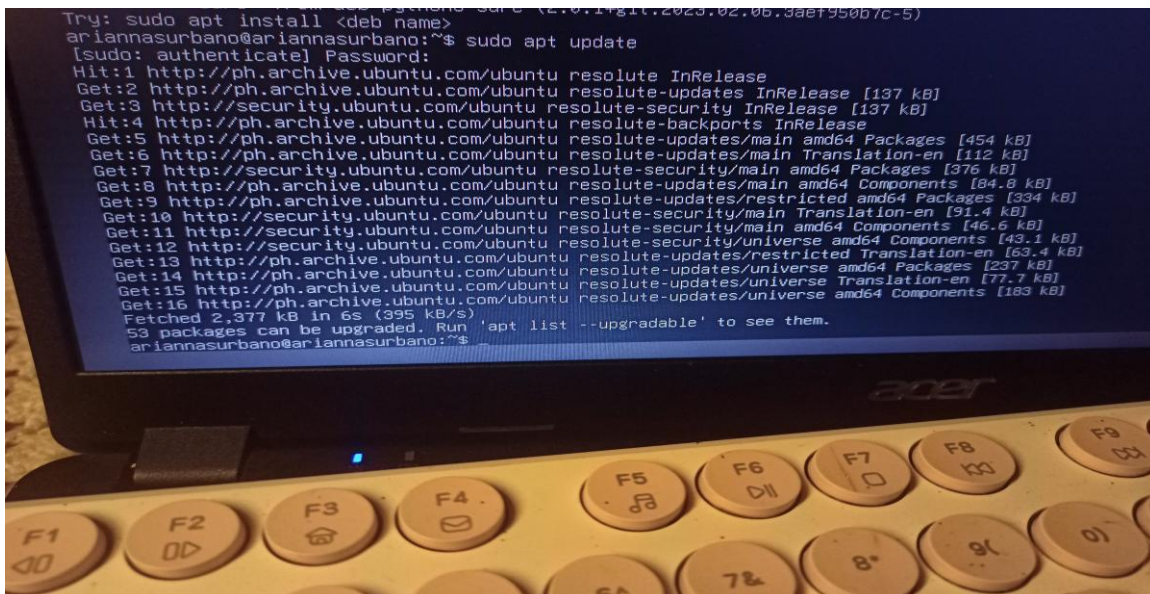
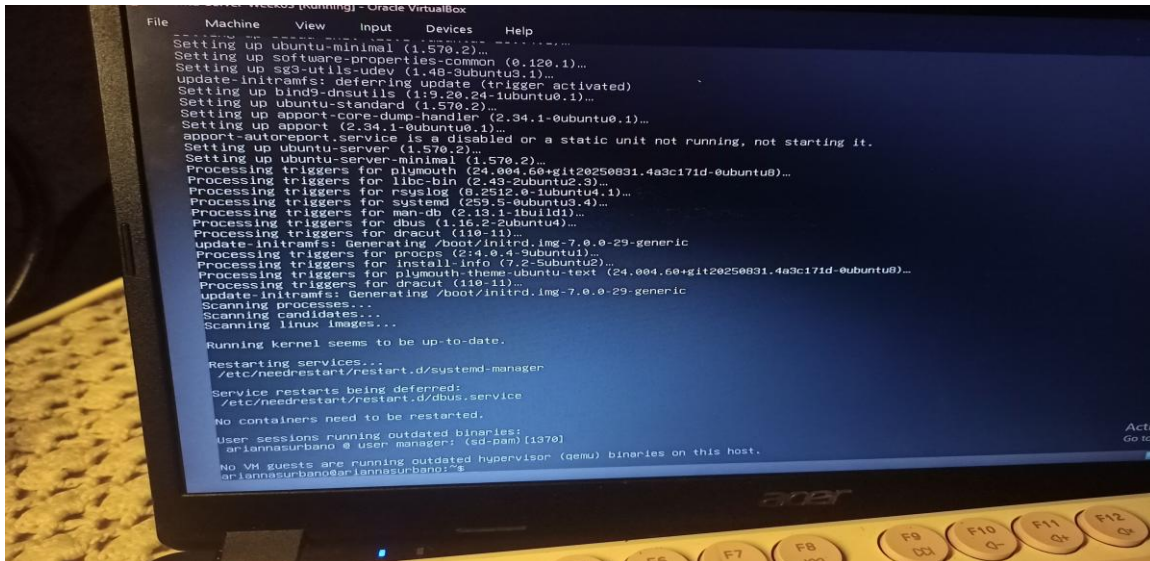
3.4 Update the Server

Run:

`sudo apt update`

`sudo apt upgrade -y`

Wait until the update and upgrade process finishes.

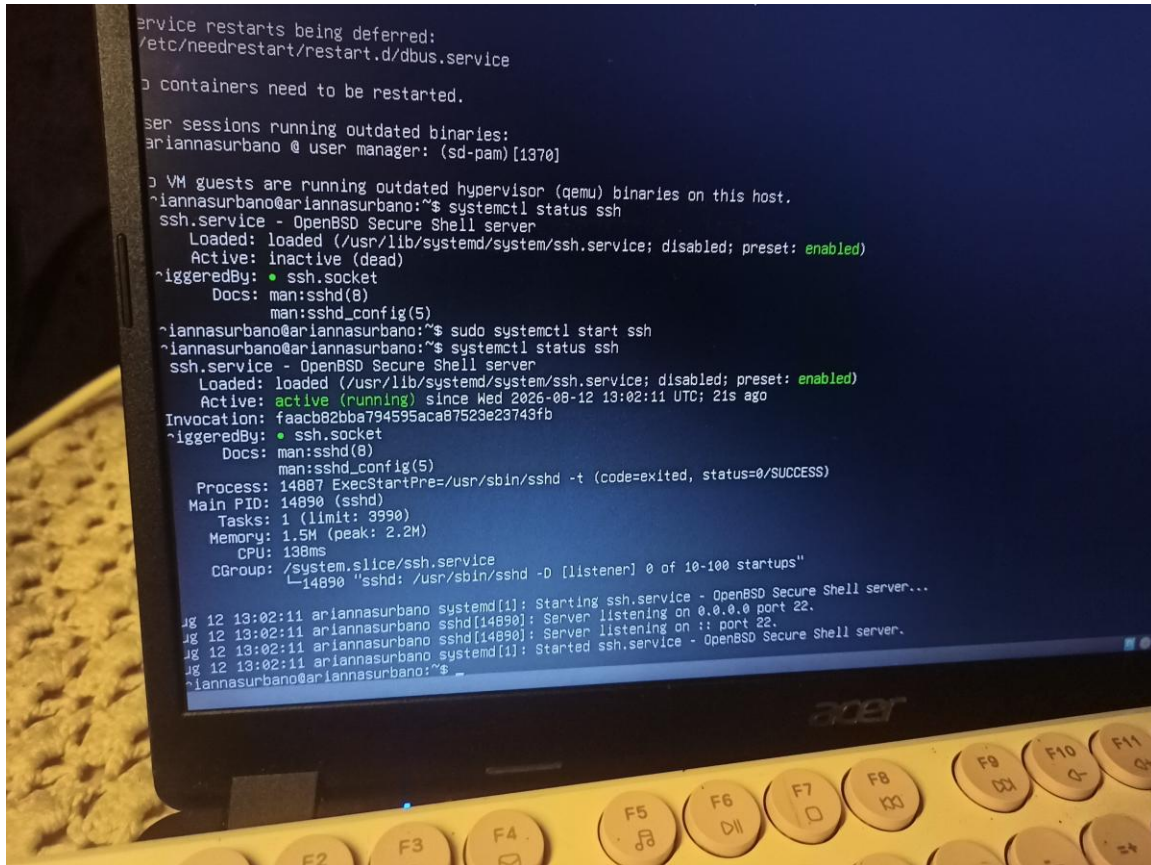


3.5 Verify SSH

Run:

`systemctl status ssh`

The service should show: active (running).



4. Server Verification Procedures

Purpose	Command	Expected Result
Hostname	<code>hostname</code>	ariannasurbano
IP Address	<code>ip addr</code>	10.0.2.15
Internet	<code>ping -c 4 google.com</code>	Successful replies
Update	<code>sudo apt update</code>	Successful update
Upgrade	<code>sudo apt upgrade -y</code>	Successful upgrade
SSH	<code>systemctl status ssh</code>	active (running)

These checks confirm that the server is installed, connected to the network, updated, and running the SSH service.

5. Troubleshooting Guide

5.1 Package Update Error

Problem: The Ubuntu server encountered an error while accessing the package repository.

Solution: Check the configured Ubuntu repository and change it to a working Ubuntu mirror if necessary. Then run:

```
sudo apt update  
sudo apt upgrade -y
```

5.2 Windows Server VM Failed to Boot

Problem: The Windows Server virtual machine failed to boot because the installation ISO was not properly mounted.

Solution:

1. Power off the VM.
2. Open VirtualBox → Settings.
3. Select Storage.
4. Check the Optical Drive.
5. Select the correct Windows Server Evaluation ISO.
6. Save the settings.
7. Start the VM again.

5.3 SSH Service Not Running

If SSH does not show active (running), check the service:

```
systemctl status ssh
```

If necessary, start it with:

```
sudo systemctl start ssh
```

Then verify again:

```
systemctl status ssh
```

6. Best Practices

1. Use strong passwords for server accounts.
2. Keep the operating system updated.
3. Verify network settings after installation.
4. Check important services using `systemctl status`.
5. Use SSH securely when remote administration is required.
6. Keep documentation and screenshots after major configuration steps.
7. Keep backups of important server data and configurations.
8. Do not share administrator passwords in documentation or screenshots.
9. Review error messages carefully before changing configurations.
10. Use only trusted software repositories when installing packages.

7. References

- Ubuntu Server Documentation — installation, configuration, and server administration
[Ubuntu Server Documentation](#)
- Ubuntu Server Reference — commands, system requirements, and technical information
[Ubuntu Server Reference](#)
- Microsoft Learn – Windows Server — Windows Server documentation
[Microsoft Learn](#)
- Microsoft Windows Server Evaluation Center — Windows Server Evaluation ISO
[Windows Server Evaluation Center](#)
- UEFI Forum — official UEFI specifications
[UEFI Forum Specifications](#)
- diagrams.net (Draw.io) — flowchart creation
[diagrams.net / Draw.io](#)